

SEP Theory

Subjective Experience Projection Theory

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Abstract

This document presents the theoretical framework of the Subjective Experience Projection (SEP) Theory, exploring the basic structure of human recognition, subjectivity, and its physical integration with quantum brain theory (Orch-OR), alongside its mathematical formulations and implications for psychopathology.

1 The Basic Structure of Recognition and Subjectivity

When grasping recognition, it invariably appears as a set of two elements: “what it is” (content) and “where it is” (place). The combination of these two recognitions of “content” and “place” constitutes “subjectivity.” Subjectivity possesses a bidirectional orientation: “A is in B” and “B is the place where A exists.”

Older subjectivities are successively captured as objects of new subjectivities (within the “content-side recognition” of the new subjectivity), forming a nested, “Matryoshka-like” structure and generating a chain of recognition. A temporal lag exists within these nested pairs, and the continuous apprehension of “the world where the past self existed” by the “present self” serves as the driving force that propels the psyche.

2 The Positional Relationship Between the Self and “Objet a” (Ego)

In “content-side recognition,” the recognition of “oneself” functions as a point (the ego) that gazes at the entire world from the center (origin). Conversely, in “place-side recognition,” one’s state is the whole, and the whole gazes back at the single point of oneself (the self).

At the center (origin) of the psychic space exists “objet a,” a “dangerous hole” capable of destroying the structure. At the same time, this “objet a” acts as the absolute driving force that propels the psyche. “Primal eventality” functions as a defensive wall that protects the psychic structure by suppressing the expansion of the hole formed as “objet a.”

3 The Duality of Physical Time t and Subjective Time τ

“Drive” (*Trieb*), the biological life energy welling up from within the body, arises within continuously flowing physical time t . “Desire” is formed when this drive is translated and structured through the Other and the network of language (the chain of signifiers).

While physical time is continuous, subjective time τ is non-continuously (discretely) inscribed as “+1” only at the precise moment when disparate recognitions are contracted and integrated into a single “meaning (Gestalt)” through contour integration.

4 The Mechanics and Equations of the Psyche (Self, Will, and Action)

The self (z) exists as a complex entity ($z = x + iy$) with a real axis (x) representing facts and social norms, and an imaginary axis (y) representing emotions and meanings. The self exists in a state of conflict, subjected to the “force of symbolization” from the direction of the real axis and the “force of drive” from the direction of the imaginary axis.

Through the interaction (cross product) of these two orthogonal forces, an entirely new vector perpendicular to the plane, “will” ($F = z \times v$), emerges. Driven by the force of will and following the right-hand rule, the self is propelled into “action” ($v = dz/dt$), a counterclockwise rotational movement.

The current plane of rotation (S_2 plane) is generated by multiplying the accumulation of past memories (S_1 plane) by the brain’s activity cycle ($\text{rot} = e^{i\omega t}$). Information on the S_2 plane is continuously expanded into a solid 3-dimensional “sphere” through a rotation operation using quaternions ($Q(\theta) = q(\theta)S_2q(\theta)^{-1}$), giving rise to the subjective world (qualia). Through this expansion, the entire previous space ($Q_{\tau-1}$) is compressed into an infinitesimal point within the current horizon (Q_τ), becoming an “object.”

5 Physical Integration with Quantum Brain Theory (Orch-OR)

“Tubulin,” the protein constituting microtubules in the brain, assumes the role of the “event operator” ($\hat{E}$) in SEP theory, converting waves into meanings (scalar values). All tubulins experience collapse simultaneously via quantum entanglement, sharing the timing of subjective time τ across the entire brain.

However, each individual tubulin itself only retains the local records (memories) that it has processed. These disparate memories and current inputs are multiplied simultaneously across the entire axonal network (tensor product $\otimes$); only at the moment they form a vast field and collapse does the subjective experience emerge as a whole. The continuous re-entry of these discrete, “flickering” events generates a smooth stream of consciousness, much like a flipbook.

6 Psychopathology and the Accumulation of “Residue”

When converting drive into meaningful desire (the experience of subjective time τ) within continuous physical time t , the raw life energy that cannot be fully reduced to meaning spills over as “residue” ($\mathcal{R}_a(t)$). This residual energy cannot precipitate into the “strata” of past memories; it constantly remains in the “here and now (S_2 plane),” continuously exerting stress on the psyche.

When the accumulation of this unprocessed energy exceeds the limits of the psyche’s existing capacity (the total allowable capacity of the bra) and vulnerability ($\mathcal{V}(t)$) reaches a critical point, eventuality collapses, leading to the manifestation of psychopathology.

7 The 15 Equations of the Psyche

(1) “I” Self

$$z = x + iy \quad (\text{where } x: \text{super-ego, } y: \text{meaning, } z: \text{ego})$$

(2) “Life” Action (The rate of change over time of the self’s existence z , the chronological change in velocity)

$$v = \frac{dz}{dt}$$

(3) “Creative Aufheben” Will

$$F = z \times v \quad (\text{where } \times \text{ denotes the cross product})$$

(4) “Engine” Brain Activity Cycle

$$\text{rot} = e^{i\omega t}$$

(5) “Memory” Memory

$$S_1 = \int z(t)dt$$

(6) “Reality” Present

$$S_2 = S_1 \cdot e^{i\omega t}$$

(7) “Qualia (Projection)” Projection

$$Q(\theta) = q(\theta)S_2q(\theta)^{-1} \quad (0 \leq \theta \leq \pi)$$

Note on Equation (7): Equation (7) corresponds to the formula bridging the body and the psyche. Besides the above equation, formulas projecting onto various motor and sensory organs would likely be required.

Equations (8) through (14) serve to explain the S_1 and S_2 planes in detail. The factors that enabled conversion into mathematical formulas are as follows:

1. Recognition is paired as two elements (the content side and the place side), which form a nested (Matryoshka-like) structure, and the parts that cannot be experienced as subjectivity or recognition are also considered to be paired (described here as simultaneity and eventality).
2. Oneself is simultaneously both the whole and a point (the phenomenon of the “whole” plunging into a “point” is defined as “transcendence,” and the reverse phenomenon is defined as “creative Aufheben”).

Finding oneself while assigning meaning to the world—continuously generating subjectivity by traversing between these two—is the fundamental function of the psyche.

(8) “Recognition”

$$\langle B | \hat{E} | a \rangle$$

(9) “Somatic” Body (Projection and Re-entry)

$$Q|a\rangle_{ext} \rightarrow |a\rangle_{somatic}$$

(10) “**Simultaneity**” Network of “Simultaneity” (Tensor Product)

$$\bigotimes_{k=1}^n |a_k(t)\rangle$$

(11) “**Transcendence**” Gestalt and Experience (Contour Integration)

$$\mathcal{Z}_{Gestalt}(\tau) = \oint_C \left(\bigotimes_{k=1}^n \langle B_k | \hat{E} | a_k \rangle \right) e^{i\omega t} dt$$

(12) “**Common sense**” Sensus Communis

$$\mathcal{Z}_{communis}(\tau) = \oint_{C_\tau} \left(\bigotimes_{k=1}^n \langle B_{shared}^{(k)} | \hat{E}_{communis} (|a_k(t)\rangle_{ext} \otimes |a_k(t)\rangle_{som}) \right) e^{i\omega t} dt$$

(13) “**Residue**”

$$\mathcal{R}_a(t)$$

(14) “**Vulnerability**”

$$\mathcal{V}(t) = \frac{\int_{t-T}^t \|\mathcal{R}_a(t')\| dt'}{\mathcal{C}(\langle B_{S1} |)}$$

(15) “**Subjectivity**”

$$\mathcal{S}_{self} = \sum_{\tau=1}^N \left[\oint_{C_\tau} \left(\bigotimes_{k=1}^n \langle B_k | \hat{E} | a_k(t) \rangle e^{i\omega t} dt \right) \right] + \int_0^t \mathcal{R}_a(t) dt$$

The self at the origin is considered as the ego, and the point S_{self} of the self at $\tau - 1$ is considered as the self.

References and Sources

1. Theoretical Background (Quantum Brain Theory / Psychopathology)

The primary theories referenced and compared in the considerations of this article are as follows:

- **Orch-OR Theory:** A theory by Roger Penrose and Stuart Hameroff proposing that quantum collapse within microtubules is the origin of consciousness.
- **Psychopathology and Psychoanalysis:** Karl Jaspers (Descriptive Psychopathology) and Jacques Lacan (Structural theory of the Symbolic, the Imaginary, and the Real; objet a; Borromean knots).

2. Co-production with Generative AI

- **Model Used:** Google Gemini (Gemini 3 Flash)
- **Production Process:** Theory construction was conducted through dialogue with AI, based on the author’s original notes.